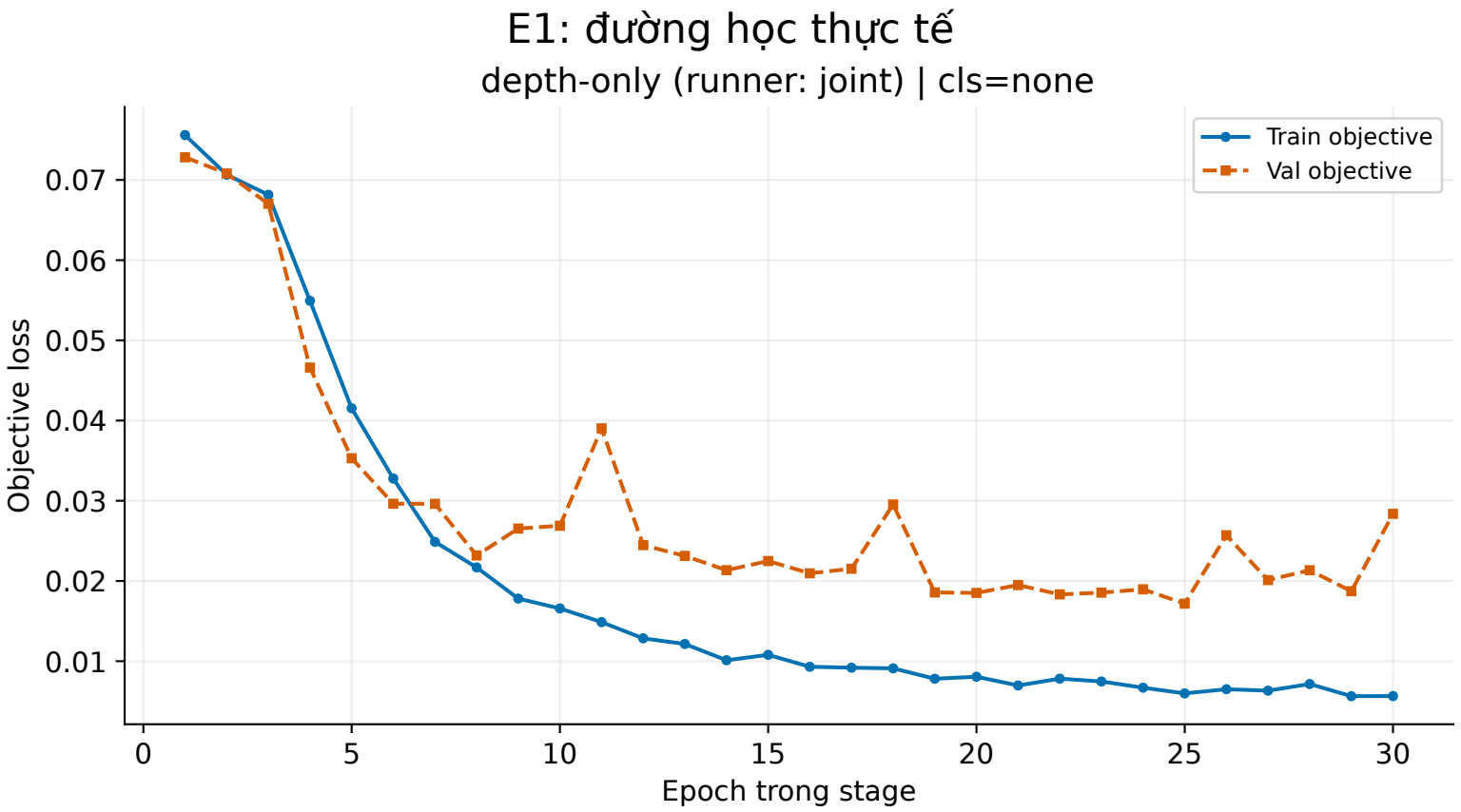
E1: đường học thực tế
depth-only (runner: joint) | cls=none
Train objective
Val objective
Objective loss
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
5
10
15
20
25
30
Epoch trong stage
CASIA development protocol · seed 42 · video mean of 20 frames · 270 attack / 90 live test videos. Raw unsmoothed logs. Stage
objectives differ: panels are separate and their losses are not comparable across stages/models.